

# LLM-Based Coordination for Heterogeneous Multi-Robot Warehouse Exploration

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## INTRODUCTION

**Warehouses are complex environments** with cluttered layouts and objects at multiple heights, requiring different robot types to collaborate effectively. **Traditional multi-robot systems** use rigid communication pipelines or task-specific planners → **less flexible** in dynamic settings.

**LLMs enable robots to:**

- Share **high-level semantic observations**
- Communicate via **natural language**
- Reason about tasks and **request assistance**
- Coordinate more adaptively

**In our system:**

- A **UGV, drone, and humanoid robot** run onboard LLM modules
- Robots exchange observations, plan collaboratively, and assist each other
- A **scene graph** provides structured knowledge of objects and spatial relationships

**Simulated warehouse experiments show:**

- Faster and more efficient exploration
- Higher accuracy
- Improved coordination vs. non-collaborative baselines

Our objective is to evaluate how LLM-based reasoning and scene-graph sharing improve coordination in heterogeneous multi-robot warehouse exploration.

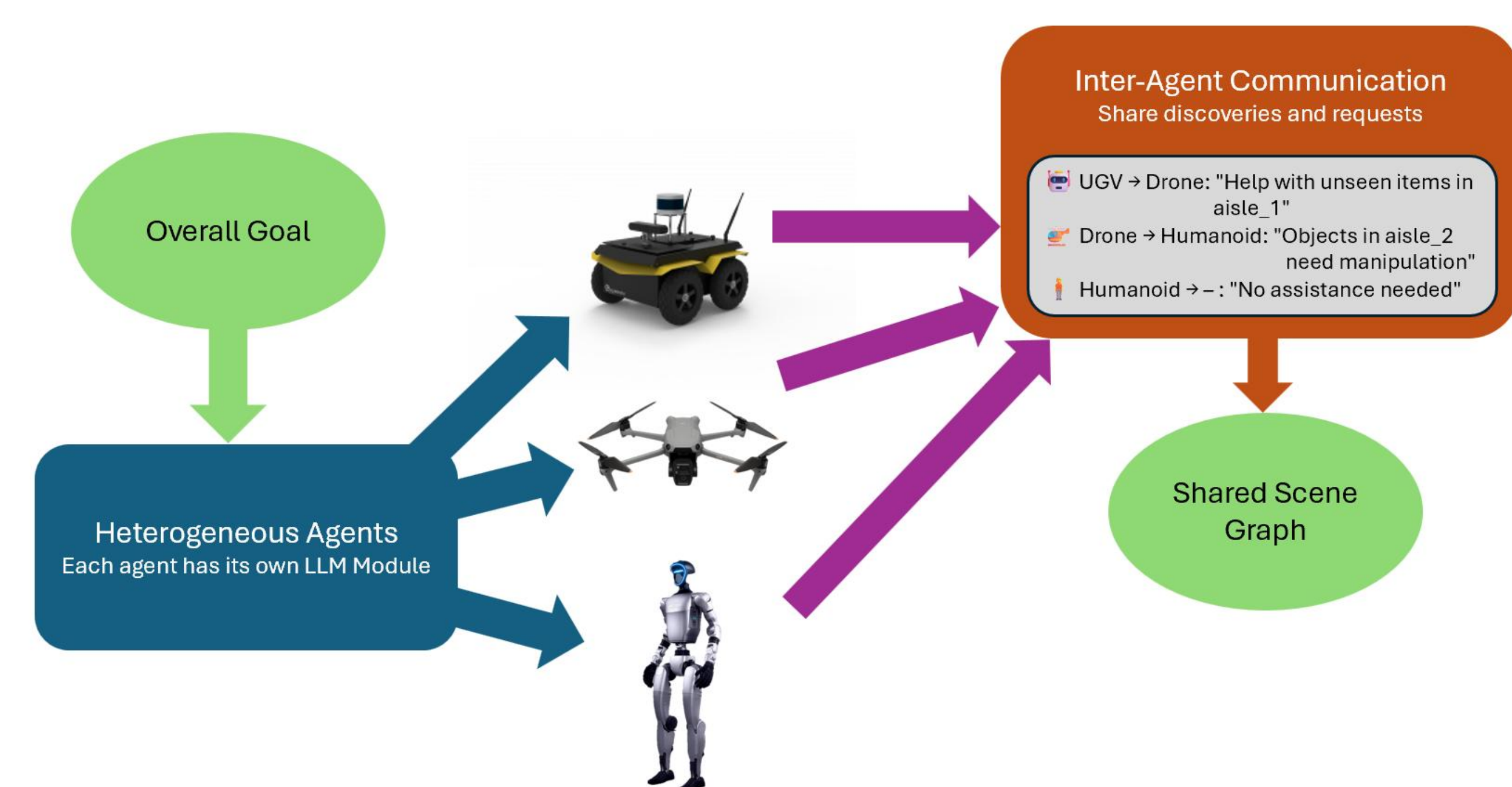


Fig. 1. Overview of the LLM-guided multi-robot warehouse exploration framework. The UGV, drone, and humanoid robot communicate via LLMs to share observations and request assistance.

## APPROACH

**Team:** UGV, Drone, Humanoid

**Initial Allocator:** Central allocator assigns aisles + observations

**Coordination:** Decentralised, via LLM messages

**Robot LLM Modules:** Parse observations + request assistance

**Scene graph:** Share updates for joint reasoning

## RESULTS

We evaluate our framework on 5 randomly generated warehouse datasets, each with 100 aisles and 2 objects per aisle (Fig. 2).

• **Experiment 1 - Full Exploration:** Robots explore until all objects are added to the scene graph. The metric is the number of aisles traversed.

• **Experiment 2 - Partial Exploration:** Each robot stops after traversing 50 aisles, simulating limited battery or resource constraints. The metric is exploration accuracy.

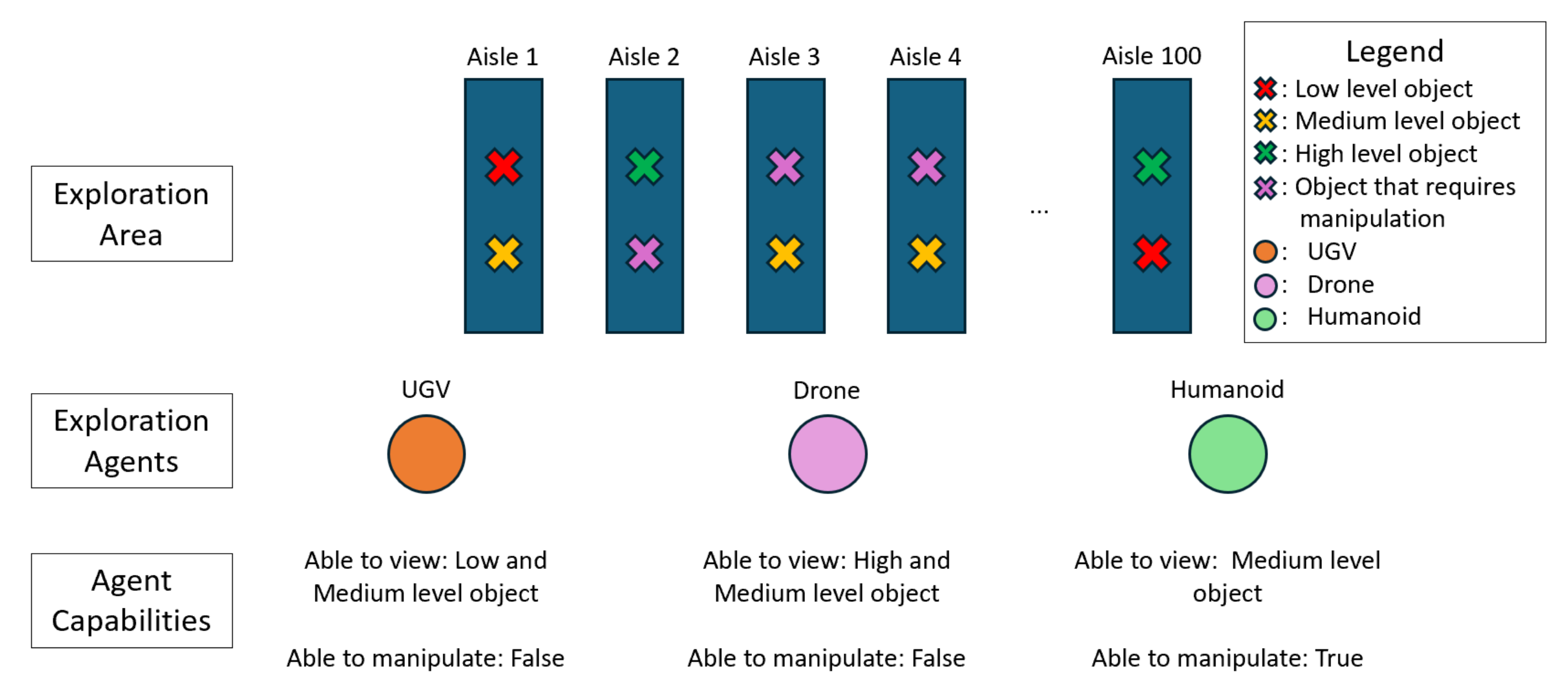


Fig. 2. Simulated warehouse experimental setup with 100 aisles. Robots traverse aisles to identify and manipulate objects.

TABLE 1  
COMPARISON OF EXPLORATION PERFORMANCE ACROSS EXPERIMENTS

| Condition               | Full Exploration<br>(Avg. Aisles Traversed) | Partial Exploration<br>(Avg. Accuracy) |
|-------------------------|---------------------------------------------|----------------------------------------|
| LLM-based Collaboration | 190                                         | 78.4%                                  |
| No Collaboration        | 300                                         | 63.5%                                  |

## CONCLUSION

LLM-guided communication enables flexible coordination of heterogeneous robots, improving efficiency and accuracy over non-collaborative baselines and highlighting the potential of language-driven multi-robot collaboration in logistics.

## ACKNOWLEDGEMENTS

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